

Benchmarking ADAPT-VQE and HI-VQE at 12 and 24 Qubits

Verified reproductions with classical controls
in the same active space

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Abstract

Two published quantum-chemistry benchmarks are reproduced under a common verification protocol. The first is imipramine in CAS(6e,6o)/6-31G, 12 qubits, treated with ADAPT-VQE. The second is Li₂S in CAS(12e,12o)/STO-3G, 24 qubits, 853,776 determinants, treated with HI-VQE along the Li–S dissociation coordinate. Each Hamiltonian carries a validation receipt and a SHA-256 fingerprint of the code that produced it, and every reported quantity is traceable to those fingerprints.

The numerical results are satisfactory. ADAPT-VQE converges to 0.017 mHa of the exact active-space energy for imipramine, and HI-VQE reproduces exact diagonalisation to better than 0.15 mHa at eleven of thirteen Li₂S geometries while sampling 2.3% of the determinant space and requiring a single measurement setting in place of 15,697 Pauli words.

The principal contribution, however, concerns the validity of the comparisons themselves. Three of the study’s headline quantities were found to compare states that were not comparable, and each failure took the same form. The two ADAPT calculations, presented as a comparison of operator pools, are shown to follow an identical trajectory for 31 iterations and to differ only in operator budget. The CASC reference at 3.20 and 3.60 Å is shown not to be the ground state of the symmetry sector it was intended to diagonalise, by two independent arguments: the variational bound satisfied by HI-VQE, and the non-monotonicity of the reference potential-energy curve. The classical baselines beyond 4.20 Å are constructed on a closed-shell singlet reference while the reference against which they are scored is a triplet, so the apparent degradation of CCSD(T) from 0.27 to 29.83 mHa coincides with a change of spin sector rather than with any discontinuity in the correlation diagnostic.

Two positive conclusions follow. First, the detection of the invalid reference is not incidental but systematic: a method returning a rigorous upper bound on the ground state of a symmetry sector constitutes a *certificate* on any reference purporting to be that ground state, so a variationally bounded sampler can audit the classical calculation against which it is being scored. Second, the reduction in measurement cost from 15,697 Pauli words to a single computational-basis setting is structural, independent of random seed, and unaffected by every comparison questioned above; it is the one quantum-side advantage established here without qualification.

The surviving claims are reported, the claims that these data cannot support are stated explicitly, and the experiments that would decide each are identified.

Guide to the reader

Interest	Sections
Summary only	Section 1.4, the five principal results, and Table 1.
Assessment of validity	Sections 5 and 6, read together. The second states the cost of the first.
ADAPT-VQE practice	Section 3, in particular Section 3.5, which bears on how operator-pool comparisons should be reported.
SQD and HI-VQE practice	Section 4, in particular Sections 4.4 and 4.5, which describe two failure modes that produce plausible numerical output.
Reproduction	Appendices A–D. Every quantity reported here is addressed by a fingerprint listed there.

Conventions. Energies are given in hartree (Ha) and errors in millihartree (mHa); $1.6 \text{ mHa} \approx 1 \text{ kcal mol}^{-1}$ is taken as chemical accuracy. Error denotes $E_{\text{method}} - E_{\text{reference}}$, the reference being exact diagonalisation of the identical active-space Hamiltonian. A positive error is variationally expected; a negative error requires explanation, and is given one in Section 4.4. The term *exact* is used throughout in the restricted sense of exact within the stated active space and basis.

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1 Introduction

1.1 Motivation

A quantum-chemistry benchmark reports the difference between two energies. That difference is meaningful only if the two energies describe comparable states: the same Hamiltonian, the same particle number and, most easily lost, the same spin sector. Where this condition fails, the arithmetic remains well defined and a number of the expected magnitude is still produced.

This is not a hypothetical concern. In the course of two routine reproduction studies the condition failed three times, in three distinct places, and in each instance the workflow produced a plausible quantity that would have been reported as a result:

1. a comparison of two operator pools in which the two calculations were, for the duration of the shorter run, the same calculation (Section 3.5);

2. a reference energy that was not the ground state of the symmetry sector it was intended to diagonalise (Section 4.4);
3. classical baselines scored against a reference lying in a different spin sector from their own (Section 4.5).

Each was identified by instrumenting a quantity other than the energy. None would have been identified by an error estimate on the energy alone.

1.2 Systems studied

The two systems were selected to span the correlation regime, and both have published counterparts that fix the active space, so that the active space is not a free parameter in this work.

- **Imipramine, CAS(6e,6o)/6-31G, 12 qubits** [2], a tricyclic antidepressant comprising 152 electrons in 237 spatial orbitals, of which six electrons in six frontier orbitals are correlated. The active space is weakly correlated, containing 20.7 mHa of correlation energy with the Hartree–Fock determinant dominant throughout, and is small enough that the exact solution is a 400×400 diagonalisation.
- **Li₂S, CAS(12e,12o)/STO-3G, 24 qubits** [1], a lithium–sulfur battery species examined along a bond-dissociation coordinate. At extended Li–S separation the reference determinant ceases to dominate, natural occupancies become fractional, the T_1 diagnostic exceeds 0.12, and the ground state changes spin sector. Each failure mode described in this work appears somewhere along this coordinate.

Both are treated under the single protocol of Section 2.

1.3 Scope and claims

This is a reproduction study with controls rather than a methods development paper. Published algorithms are implemented on the active spaces specified by their source publications and assessed against exact diagonalisation and against classical baselines computed here.

No claim is made to new algorithms; to hardware results, all calculations being CPU simulations; to reproduction of the source publications’ total energies, which depend on geometries and orbital sets those publications do not report; to statistical significance, the stochastic calculations having been performed at a single random seed; or to extrapolation beyond 24 qubits, where verification by exact diagonalisation is no longer possible.

Section 6 enumerates the claims declined. It is a component of the result rather than a list of qualifications appended to it.

1.4 Principal results

Result 1 The two ADAPT calculations follow an identical trajectory for 31 iterations

The UCCSD-pool and UCCGSD-pool calculations select the same 31 operators in the same order, with energies and pool gradients agreeing to better than 10^{-12} at every step. The UCCSD calculation terminated on a 31-operator budget with a pool gradient of 2.05×10^{-2} remaining, twenty times its own convergence threshold. The interval between 0.892 mHa and 0.017 mHa is therefore attributable to truncation rather than to pool composition, and no conclusion regarding UCCSD relative to UCCGSD is drawn (Section 3.5).

Result 2 The observed spin contamination is a truncation artefact

Since the trajectories coincide, the value $\langle \hat{S}^2 \rangle = 2.8 \times 10^{-3}$ recorded at 31 operators is a property of the truncated state and not of the UCCSD generator set. Continuation of the same trajectory to 58 operators reduces it to 1.1×10^{-6} . Symmetry breaking measured before convergence is therefore not evidence that a pool cannot represent the symmetric state.

Result 3 Two reference energies are invalid, by two independent arguments

At $r = 3.20$ and $3.60 \text{ \AA}$, HI-VQE returns energies 8.50 and 24.24 mHa below CASCI. Every HI-VQE energy is the lowest eigenvalue of *PHP* and hence an upper bound on the ground state of the sampled $S_z = 0$ sector, so the reference cannot be that ground state. Independently, the reference curve is non-monotonic across these geometries, rising from -407.8535 at $3.20 \text{ \AA}$ to -407.8254 Ha at $3.60 \text{ \AA}$ and then falling to -407.8407 Ha at $4.20 \text{ \AA}$, while the HI-VQE curve increases monotonically to the asymptote. All errors at these two geometries, for every method, are void (Section 4.4).

Result 4 The degradation of CCSD(T) coincides with a change of spin sector

CCSD(T) errs by 0.27 mHa at $3.60 \text{ \AA}$ and by 29.83 mHa at $4.20 \text{ \AA}$. Across this interval the T_1 diagnostic increases smoothly from 0.0896 to 0.1134, whereas the reference changes from $\langle \hat{S}^2 \rangle = 0$ to $\langle \hat{S}^2 \rangle = 2.00$. An RHF-based CCSD(T) treatment is constrained to the singlet. The reported error is therefore a sum of a spin-sector difference and a correlation-treatment deficiency, and at 4.20 and $5.00 \text{ \AA}$ these contributions cannot be separated from these data (Section 4.5).

Result 5 At the geometry where a control exists, the classical component alone is sufficient

At $r = 2.10 \text{ \AA}$ the complete method yields 0.076 mHa; the classical selected-CI control, containing no circuit, yields 0.141 mHa; the circuit without the classical expansion yields 32.74 mHa from 304 determinants. The first two lie an order of magnitude inside chemical accuracy and the difference between them is not reported as a result, each being a single realisation of a stochastic procedure. The third is worse by a factor of 230, which exceeds any plausible seed-to-seed variation (Section 4.7).

Result 6 The reduction in measurement cost is unconditional

A conventional VQE requires 15,697 distinct Pauli words per energy evaluation on the Li_2S Hamiltonian; HI-VQE requires one computational-basis setting, because it consumes configurations rather than expectation values. This factor depends on no random seed, no reference energy and no spin sector, and is therefore unaffected by every comparison questioned in Results 1 through 5. It is the one quantum-side advantage established here without qualification (Section 4.2).

Result 7 A variationally bounded method certifies its own reference

Result 3 is an instance of a general property. Any method returning a rigorous upper bound on the ground state of a symmetry sector – HI-VQE, sample-based diagonalisation, or selected-CI – provides a one-sided certificate on any reference asserted to be that ground state: an energy below the reference falsifies the reference rather than the method. The usual hierarchy is thereby inverted, the quantum calculation auditing the classical one against which it is scored. The property is inexpensive, requires no additional computation beyond the sign of the error, and is developed in Section 5.2.

Table 1: Summary of the principal quantities. Each column employs a single control throughout; controls are not interchanged within a row. Errors are against exact diagonalisation of the identical active-space Hamiltonian. Provenance is given in Appendix A.

	Case A	Case B	
	Imipramine CAS(6e,6o), 12q	Li ₂ S, CAS(12e,12o), 24q $r = 2.10 \text{ \AA}$	$r = 6.00 \text{ \AA}$
Quantum method	ADAPT-VQE	HI-VQE	HI-VQE
Quantum error (mHa)	0.017	0.076	0.009
Best classical, same space error (mHa)	CCSD(T) 0.008	CCSD(T) 0.41	CCSD(T) 29.48
Classical component alone	—	0.141	<i>not computed</i>
Hartree–Fock error (mHa)	20.736	39.5	168.2
Determinant space	400	853,776	853,776
Fraction sampled	—	2.34%	1.30%
Measurement settings	—	1 of 15,697	1 of 15,697
Reference $\langle \hat{S}^2 \rangle$	0	0	1.947
Sector match with baseline	yes	yes	no (degenerate)

The third column is the only one in which the quantum method surpasses the best classical method available in the same active space, and the final two rows indicate why Section 4.5 is required before that column can be interpreted.

2 Methodology

2.1 Active-space construction

Both studies follow a common sequence: a restricted Hartree–Fock calculation in the full basis, selection of an active space, and reduction to a qubit Hamiltonian under the Jordan–Wigner transformation. The active space is fixed in each case by the source publication.

The step of consequence, and the one most often left implicit, is verification that the selected orbitals are those intended. For imipramine, each of the six frontier orbitals carries a Mulliken population above 0.93 on the tricyclic system, against required minima of 0.80 for occupied and 0.65 for virtual orbitals. For Li₂S, the five frozen orbitals are confirmed to constitute the sulfur 1s2s2p shell: the smallest core population on sulfur is 1.000 and the core–active orbital-energy gap is 3.88 Ha at equilibrium, while no other orbital in the molecule lies below -3 Ha . Construction is aborted if either test fails.

2.2 Provenance and validation protocol

Each cached Hamiltonian is accompanied by a validation receipt recording:

- `spec_sha256`, a digest of the molecular specification, preventing an altered molecule from reusing a cache;
- `physics_sha256`, a digest of only those modules capable of altering a Hamiltonian, so that modification of a simulator cannot invalidate a Hamiltonian constructed and validated elsewhere;
- `workflow_sha256`, a digest of every module in the workflow directory, recorded for provenance and never used to gate execution;
- the validation checks themselves: symmetry of the one-electron integrals, the eight-fold symmetry of the two-electron integrals, core–active separation, active-space dimension, and agreement

between the Hartree–Fock determinant energy evaluated from the cached integrals and from the mean-field object.

The protocol is procedural rather than scientific, and is described here for a specific reason: it established that an earlier Li₂S dataset, carrying identical file names, an identical directory structure and plausible values, had been generated under a different `physics` digest and could not be combined with the present one. The two sets differ by 10.1 mHa in the CASCI energy at 6.00 Å. In the absence of the digests they are indistinguishable (Appendix E).

Reproduction of digests across platforms

The digests are computed over raw file bytes. A repository obtained on Windows with `core.autocrlf` enabled carries CRLF line endings and therefore yields different digests from identical source on Linux. Verification of a published digest requires prior normalisation of line endings.

2.3 Acceptance criteria

Two criteria are applied and both must be satisfied.

Energy. The error against exact diagonalisation of the same active space must lie below 1.6 mHa.

Symmetry. An energy that is correct through cancellation of errors is not correct. For Case A a state is accepted only if its spin contamination is priced below 0.16 mHa, one tenth of the energy criterion, the contamination energy being taken as $\langle \hat{S}^2 \rangle$ at a price of 1 Ha per unit. For Case B the corresponding requirement is agreement between $\langle \hat{S}^2 \rangle$ of the returned state and of the reference; where these disagree the calculation is labelled `SPIN CONTAMINATED` or `INCONSISTENT` and is not reported as an error.

The second criterion is responsible for Results 1 through 4. Its cost is one additional expectation value per calculation.

2.4 Computational details

Electronic structure is computed with PySCF [9]; circuits are constructed and transpiled with Qiskit [10]; exact diagonalisation and the classical eigensolvers use SciPy sparse routines. No quantum hardware is employed. Package versions are given in Appendix C and differ between the two cases, which were executed three days apart on different images.

3 Case A: imipramine, CAS(6e,6o), 12 qubits

3.1 Molecular system and active space

Imipramine (C₁₉H₂₄N₂, PubChem CID 3696 [11]) is a tricyclic antidepressant. The geometry is a single PubChem conformer retrieved 2026-08-12 and is not one of the molecular-dynamics or nudged-elastic-band snapshots employed by the source study [2], which are unpublished. The molecule comprises 152 electrons in 237 spatial orbitals at the 6-31G level. Seventy-three orbitals are frozen and six electrons are correlated in the six canonical RHF frontier orbitals, indices 73–78, giving CAS(6e,6o), 12 qubits and a 400-determinant configuration space. The orbitals are listed in Table 2.

The qubit Hamiltonian contains 1819 Pauli terms, none discarded: the truncation threshold is 10^{−6} Ha and the L_1 bound on discarded weight is exactly zero, so the qubit Hamiltonian represents the active-space Hamiltonian without approximation.

Table 2: The imipramine active space. Orbital energies are canonical RHF eigenvalues; localisation is the Mulliken population on the tricyclic fragment.

MO index	Designation	Energy (Ha)	Localisation
73	HOMO−2	−0.32099	0.955
74	HOMO−1	−0.31702	0.948
75	HOMO	−0.26745	0.948
76	LUMO	+0.13935	0.938
77	LUMO+1	+0.14631	0.961
78	LUMO+2	+0.15199	0.938

Interpretative scope of a 400-determinant active space

CAS(6e,6o) constitutes a 400×400 eigenvalue problem, solved exactly in microseconds. Neither CCSD(T) nor a quantum algorithm would be deployed on it for its own sake. Case A is accordingly a calibration: it establishes what each method achieves where the exact result is known, and its significance depends entirely on whether the ordering it reveals persists into active spaces where the exact result is not known. That question is returned to in Section 3.7 and is treated there as a question rather than as a finding.

3.2 Validation of the active-space Hamiltonian

The validation receipt records a sector ground-state energy of -842.0180419413565 Ha with $\langle \hat{S}^2 \rangle = -4.9 \times 10^{-20}$, and a Hartree–Fock energy of -841.9973061557878 Ha. Independent exact diagonalisation yields -842.0180419413552 Ha; the two agree to 1.3×10^{-12} Ha, four orders of magnitude below any quantity reported here. All checks pass. The correlation energy available within the active space is therefore

$$E_{\text{corr}} = E_{\text{CASCI}} - E_{\text{HF}} = -20.7358 \text{ mHa}.$$

Chemical accuracy corresponds to recovery of 92% of this quantity, which is not a demanding criterion, and the results below should be read against it.

3.3 ADAPT-VQE protocol

ADAPT-VQE [3] constructs an ansatz iteratively. At each step the energy gradient of every operator in a pool is evaluated with respect to a new parameter appended at the front of the current ansatz; the operator of largest gradient is appended; all parameters are re-optimised; and the procedure terminates when the largest available gradient falls below a threshold.

Both calculations begin from the Hartree–Fock determinant with all angles set to zero, employ a gradient tolerance of 10^{-3} , and impose neither number nor spin penalty, the validation step having confirmed that the unconstrained ground state of the qubit Hamiltonian lies in the target sector with 6.0 electrons and $\langle \hat{S}^2 \rangle = 0$. Energies are obtained from exact sparse statevector algebra; the final ansatz is additionally synthesised as a Qiskit circuit and its energy recomputed, agreeing with the algebraic value to 8×10^{-10} Ha in both cases.

The two calculations differ in operator pool and in operator budget (Table 3). The second difference proves to be the operative one.

3.4 Results

The converged calculation recovers 99.92% of the available correlation energy and lies 0.017 mHa from the exact result, a factor of 93 within chemical accuracy, which it first attains at 28 operators

Table 3: The two ADAPT-VQE configurations. The operator budgets differ by more than a factor of three, which is the subject of Section 3.5.

	Calculation 1	Calculation 2
Operator pool	UCCSD	UCCGSD
Pool size	63	435
singles	9	15
doubles	54	420
Operator budget	31	100
Gradient tolerance	10^{-3}	10^{-3}

Table 4: ADAPT-VQE results for imipramine CAS(6e,6o). The exact active-space energy is -842.0180419414 Ha. Contamination energy is $\langle \hat{S}^2 \rangle$ priced at 1 Ha per unit; the acceptance budget is 0.16 mHa.

	UCCSD pool, 31 ops	UCCGSD pool, 58 ops
Energy (Ha)	-842.0171502782	-842.0180246890
Error (mHa)	0.892	0.017
Correlation recovered	95.70%	99.92%
$\langle \hat{S}^2 \rangle$	2.76×10^{-3}	1.10×10^{-6}
Contamination (mHa)	2.761	0.001
Electron number	6.000000	6.000000
Converged	no	yes
Termination	operator budget	gradient $8.6 \times 10^{-4} < 10^{-3}$
Final pool gradient	2.05×10^{-2}	8.62×10^{-4}
Matrix exponentials	9188	29817
Wall time (s)	52.4	133.0
Status	SPIN CONTAMINATED	VERIFIED

(Figure 1). The corresponding trajectory in absolute energy is shown in Figure 2.

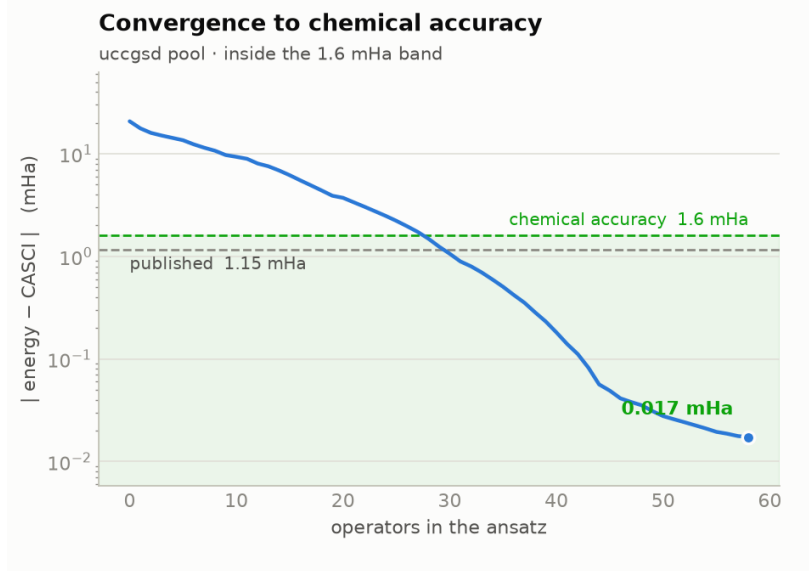


Figure 1: Convergence of ADAPT-VQE with the UCCGSD pool. The error against the exact active-space energy falls below chemical accuracy at 28 operators and terminates at 0.017 mHa. The published statevector result for the same active space, 1.15 mHa [2], is marked for reference; the two are not directly comparable, as discussed in Section 3.8.

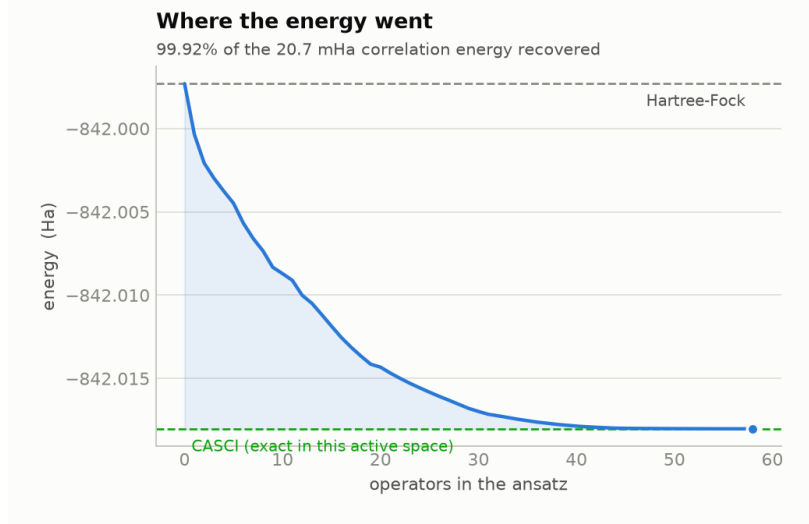


Figure 2: The same trajectory in absolute energy, from the Hartree–Fock determinant to the exact active-space energy. The shaded region is the 20.7 mHa of correlation energy available within CAS(6e,6o), of which 99.92% is recovered.

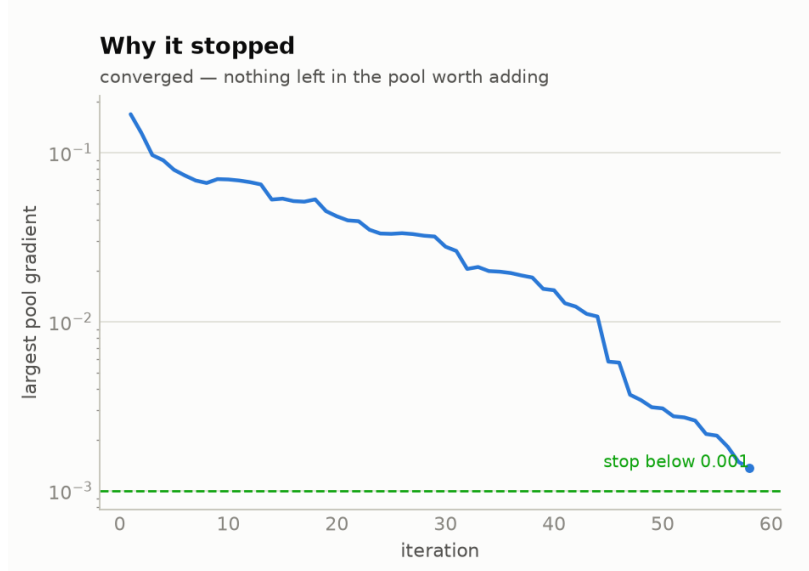


Figure 3: The largest pool gradient at each iteration, which is the quantity governing termination. The calculation stops when it falls below 10^{-3} , having exhausted the operators worth appending.

3.5 Coincidence of the two operator-selection trajectories

Table 4 invites the interpretation that the generalised pool succeeds where the standard pool fails. The data exclude that interpretation.

Comparison of the two calculations step by step shows that, for all 31 iterations performed by both, the energies and pool gradients agree to better than 10^{-12} ; they are the same values rather than similar ones. A sample of the trajectory is given in Table 5.

The explanation is direct. UCCGSD contains UCCSD; the gradient ranking selected the same operators, in the same order, from both pools; and the two calculations were consequently identical until the first exhausted its budget. The additional 372 double excitations available in the generalised pool contributed nothing to the first 31 selections.

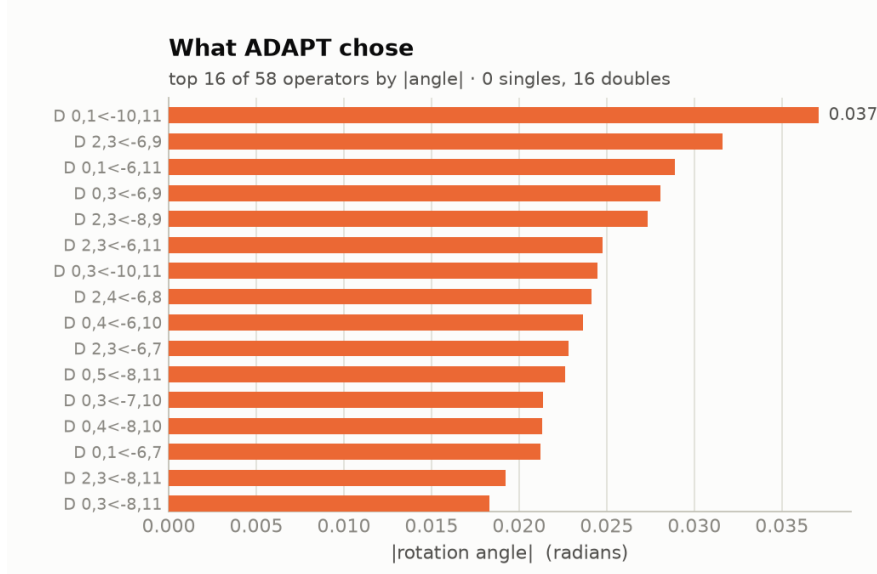


Figure 4: The sixteen largest rotation angles in the converged ansatz. All are double excitations, and no single excitation appears anywhere in either ansatz. This is expected for a closed-shell RHF reference, for which Brillouin’s theorem causes the singles gradients to vanish at the starting point.

Table 5: The first 31 ADAPT iterations of the two calculations. Operator labels differ only in the direction convention of the two pool enumerations (D 10,11 $\leftarrow$ 0,1 against D 0,1 $\leftarrow$ 10,11); the operators, energies and gradients are identical. Errors are against -842.0180419414 Ha.

Iteration	Pool gradient	Energy (Ha)	Error (mHa)
1	1.687×10^{-1}	-842.000348871	17.693
2	1.303×10^{-1}	-842.002070089	15.972
3	9.661×10^{-2}	-842.002982864	15.059
5	7.903×10^{-2}	-842.004488576	13.553
10	6.934×10^{-2}	-842.008708670	9.333
15	5.338×10^{-2}	-842.011863405	6.179
20	4.195×10^{-2}	-842.014322432	3.720
25	3.304×10^{-2}	-842.015825043	2.217
30	2.775×10^{-2}	-842.016982405	1.060
31	2.617×10^{-2}	-842.017150278	0.892
<i>Calculation 1 terminates here on its operator budget. Calculation 2 continues:</i>			
40	1.533×10^{-2}	-842.017860706	0.181
50	3.064×10^{-3}	-842.018014246	0.028
58	1.361×10^{-3}	-842.018024689	0.017

Not established: a comparison of operator pools

No conclusion is drawn regarding UCCSD relative to UCCGSD. Such a comparison requires both calculations to converge; the budgets here were 31 and 100, and the first terminated on its budget with a gradient twenty times its threshold. Reported as “UCCSD yields 0.892 mHa and UCCGSD yields 0.017 mHa”, Table 4 would support a conclusion contradicted by its own underlying data. The decisive experiment is a single change of configuration: repetition of the UCCSD calculation with a 100-operator budget. The expectation is convergence to within a few hundredths of a millihartree of Calculation 2, since 32 of its 63 operators were never offered.

The same reasoning disposes of the contamination result. Because the states at iteration 31 are the same state, the value $\langle \hat{S}^2 \rangle = 2.8 \times 10^{-3}$ observed there is not attributable to the UCCSD generator

set; it is a property of a truncated ADAPT ansatz, and continuation of the same trajectory reduces it by three orders of magnitude. The practical consequence is that a partially converged ADAPT ansatz may exhibit symmetry breaking while remaining on a trajectory toward a symmetric state, so contamination measured before convergence is not grounds for abandoning a pool.

Figure 5 records the termination of the truncated calculation, and Figure 6 its error trajectory.

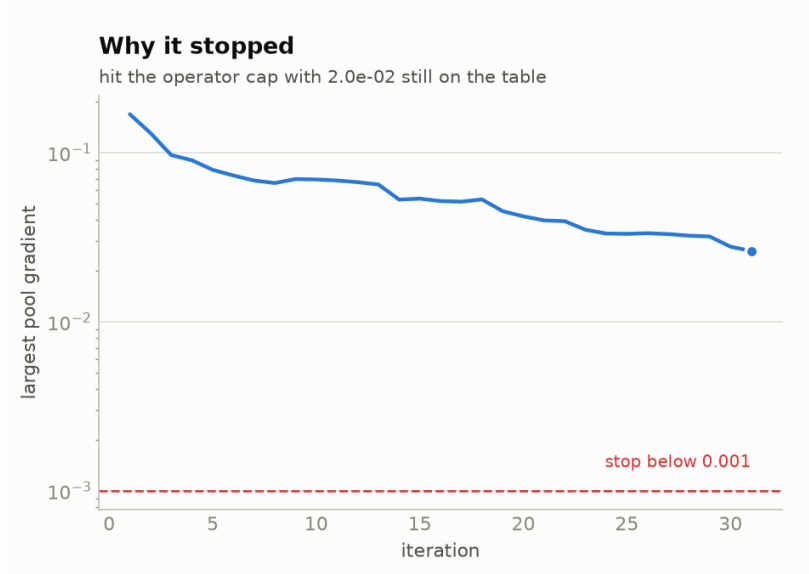


Figure 5: Termination of the truncated calculation. The pool gradient remains at 2.0×10^{-2} when the operator cap is reached, twenty times the threshold that would have justified stopping. This panel, rather than the energy, is the accurate summary of the calculation.

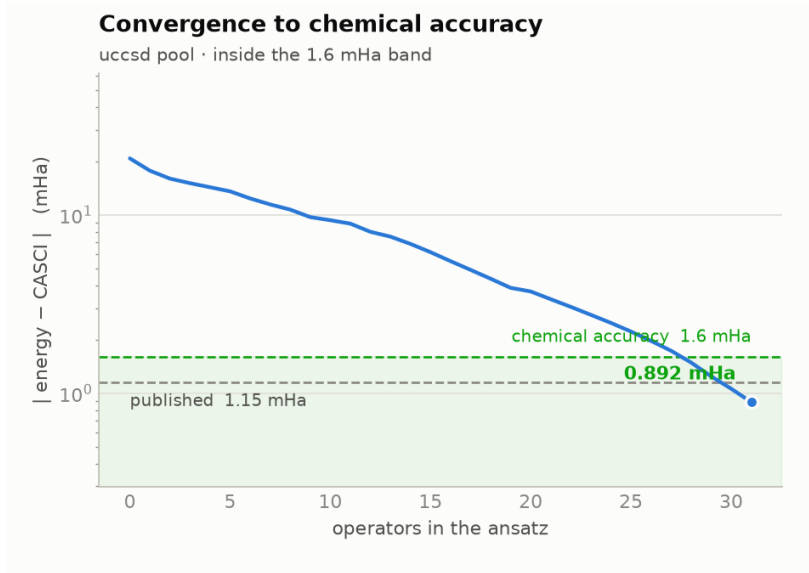


Figure 6: Error trajectory of the truncated calculation, terminating at 0.892 mHa. Over its common range this trajectory is numerically identical to Figure 1, as established in Section 3.5.

3.6 Circuit resource requirements

The converged ansatz, synthesised exactly and transpiled at optimisation level 3, requires 2988 two-qubit gates, 4674 single-qubit gates and a circuit depth of 5296 on 12 qubits, for 58 variational parameters. At two-qubit error rates presently attainable, a circuit of this depth does not return an energy of 0.017 mHa. The wall time of 133 s reported in Table 4 is accordingly simulator time

for a circuit that has no current hardware realisation, and the comparison of the following section should be read with that in mind.

3.7 Classical baselines in the same active space

MP2, CCSD and CCSD(T) were computed in the identical active space, with the identical frozen core, from the identical integrals. Results are collected in Table 6 and assessed against the criteria of Section 2.3.

Table 6: Imipramine CAS(6e,6o): all methods against the exact active-space energy. The RHF calculation common to all methods required a further 92.5 s and is excluded from every row. Wall times are indicative only; see Section 7.

Method	Energy (Ha)	Error (mHa)	Time (s)
Hartree–Fock	−841.9973061558	20.736	—
MP2	−842.0102524898	7.789	5.7
ADAPT-VQE, 31 operators	−842.0171502782	0.892	52.4
CCSD	−842.0180213368	0.021	252.9
ADAPT-VQE, 58 operators	−842.0180246890	0.017	133.0
CCSD(T)	−842.0180339214	0.008	502.1
CASCI (exact here)	−842.0180419414	0	—

CCSD(T) is more accurate than the converged quantum ansatz by a factor of two, and CCSD is comparable to it. All three lie far within chemical accuracy.

Established by Case A

In an active space for which the exact result is computable, a converged ADAPT-VQE ansatz is accurate to 0.017 mHa and is surpassed by CCSD(T). This is a calibration result: it indicates that the quantum implementation is sound and that the active space is too weakly correlated to require it.

Not established: that this ordering persists at larger scale

The question of consequence is not which method is superior on 400 determinants, but whether the ordering survives into active spaces where CASCI is unavailable. These data cannot address it. CCSD(T) is favoured here because the reference determinant dominates, and its advantage is expected to erode precisely where that ceases to hold, as it does along the Li₂S coordinate, where CCSD(T) degrades by three orders of magnitude and where ADAPT-VQE was not computed. A single weakly correlated 12-qubit result is not evidence concerning strongly correlated larger ones, in either direction. The decisive experiment is inexpensive and the implementation already exists: ADAPT-VQE on the cached Li₂S Hamiltonian at 2.10 and 6.00 Å, listed as item 2 of Section 6.

3.8 Comparison with the published study

The source study [2] reports, for imipramine CAS(6e,6o)/6-31G at 12 qubits with RHF orbitals and Jordan–Wigner mapping, a CASCI energy of −841.953249 Ha, a statevector VQE energy of −841.952101 Ha corresponding to an error of 1.15 mHa obtained with ADAPT-GQE circuits, and a hardware energy of −841.752841 Ha.

The total energies obtained here differ from those by approximately 65 mHa because a different conformer is used; the conformer of the source study is not published. Total energies are therefore not comparable and are not compared. The comparable quantity is each study’s error against its

own CASCI reference: 1.15 mHa there against 0.017 mHa here. The difference is attributed to ansatz expressiveness and convergence criteria rather than to implementation quality, the circuits of the source study having been generated under a hardware-executability constraint that did not apply here, at a resource cost documented in the preceding section.

4 Case B: Li_2S , CAS(12e,12o), 24 qubits

4.1 Molecular system and dissociation coordinate

Li_2S is treated in a linear Li–S–Li geometry with the STO-3G basis, comprising 22 electrons in 19 orbitals. CAS(12e,12o) freezes the five lowest orbitals and discards the two highest virtuals, giving 24 qubits and 853,776 determinants. The frozen set is verified to constitute the sulfur $1s2s2p$ shell, whose orbitals lie near -91 , -9 and -6.7 Ha while no other orbital lies below -3 Ha; the core–active gap is 3.88 Ha at equilibrium.

The coordinate extends one Li–S bond while the second is held at $2.10 \text{ \AA}$, over thirteen geometries from 1.70 to $6.00 \text{ \AA}$. The reference at every geometry is CASCI in the same active space.

Figure 7 indicates the character of the coordinate. At $1.70 \text{ \AA}$ the natural occupancies are near-integer. At $6.00 \text{ \AA}$ three orbitals carry fractional occupancy, approximately 1.03, 1.63 and 1.33, which is multireference character in its most direct expression.

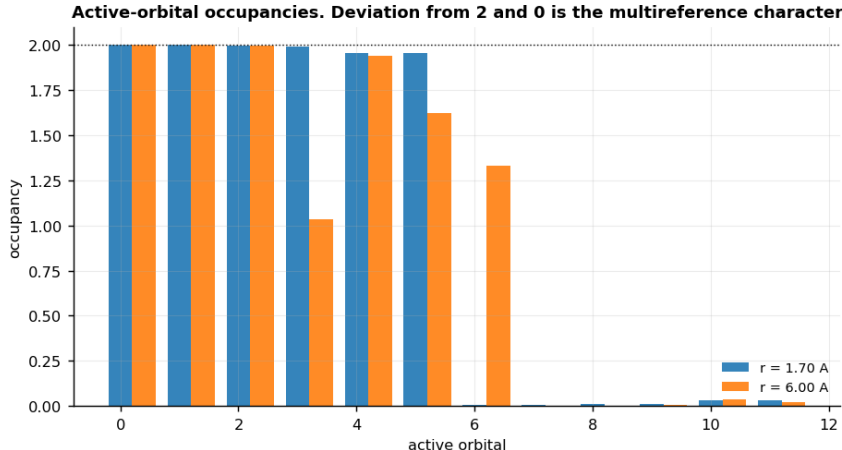


Figure 7: Active-orbital occupancies at the two extremes of the coordinate. Deviation from 2 and 0 measures multireference character. At $1.70 \text{ \AA}$ the occupancies are nearly integer; at $6.00 \text{ \AA}$ three orbitals are fractionally occupied, a configuration that a single-reference method cannot represent.

4.2 HI-VQE protocol

HI-VQE [1] replaces the expectation-value measurement of a conventional VQE [4] with a sampling step. A parameterised circuit is sampled in the computational basis; the resulting bitstrings are interpreted as electronic configurations; these, together with configurations proposed by a classical single-and-double expansion, define a subspace; the Hamiltonian is projected onto that subspace and diagonalised exactly; and the resulting energy and its gradient with respect to the circuit parameters drive the subsequent iteration. Structurally, the classical component is a selected-CI procedure in the tradition of CIPSI [5] and SHCI [6], with the circuit supplying part of the selection. This observation is returned to in Section 4.7, since it determines how demanding the classical control is.

The consequence for measurement cost is the motivation for the method. A conventional VQE applied to this Hamiltonian requires 15,697 Pauli words per energy evaluation, whereas HI-VQE

requires one measurement setting, since it requires configurations rather than expectation values (Figure 8). The circuit itself is modest by comparison with the ADAPT ansatz of Case A (Figure 9).

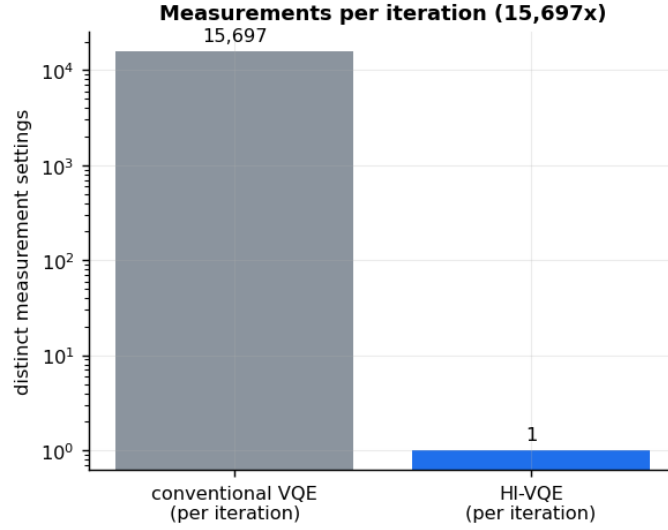


Figure 8: Distinct measurement settings required per iteration: 15,697 Pauli words for a conventional VQE on this Hamiltonian, against a single computational-basis setting for HI-VQE.

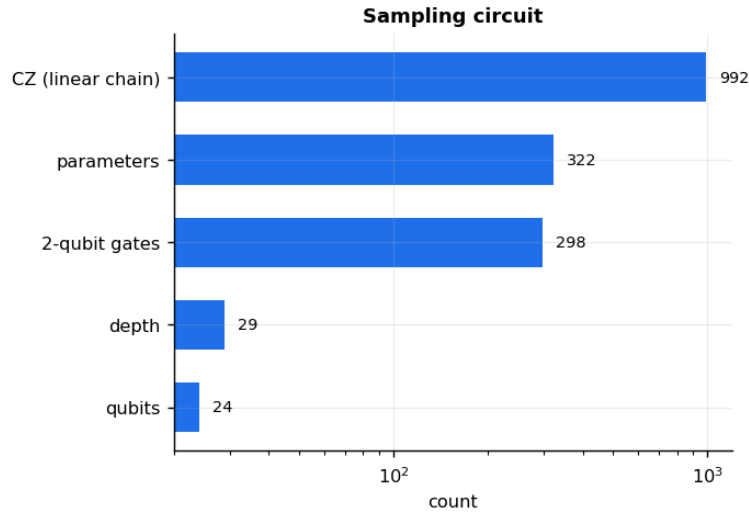


Figure 9: Resources of the sampling circuit, a hardware-efficient unitary cluster Jastrow comprising nearest-neighbour Givens rotations within each spin block and a number–number Jastrow layer coupling them. Every gate commutes with both spin-resolved number operators, so that on a noiseless simulator each shot is a valid configuration and particle-number leakage is identically zero rather than merely small.

The variational structure of the method is essential to Section 4.4. Every reported energy is the lowest eigenvalue of PHP for the projector P onto the current subspace, and is therefore an upper bound on the ground state of whichever symmetry sector the sampled configurations span, for any P . Noise in the sampler can only propose an inferior subspace; it cannot bias the energy downward.

Parameters are as follows: subspace capped at 20,000 determinants; classical expansion of 400 candidates per iteration generated from the eight leading configurations; amplitude screening at 10^{-6} ; second-order perturbative ranking; 8192 shots; one circuit repetition; SPSA optimisation; convergence at 10^{-6} Ha sustained over five iterations; random seed 1234. Complete settings are given in Appendix B.

4.3 Dissociation curve

The complete set of geometries is given in Table 7, and the curve is shown against the classical hierarchy in Figures 10 and 11.

Table 7: Li₂S dissociation, CAS(12e,12o)/STO-3G, 24 qubits. Errors are against CASCI in the same active space. The two italicised rows are void: the reference at those geometries is not the sector ground state (Section 4.4), so no method’s error there is interpretable.

r (Å)	$E(\text{HF})$	$E(\text{HI-VQE})$	$E(\text{CASCI})$	err (mHa)	HF err (mHa)	dets	% space
1.700	-407.950117	-407.985509	-407.985607	+0.099	35.5	19,992	2.342
1.900	-407.964402	-408.001987	-408.002076	+0.089	37.7	19,950	2.337
2.000	-407.960694	-407.999205	-407.999309	+0.104	38.6	19,992	2.342
2.100	-407.952748	-407.992132	-407.992208	+0.076	39.5	19,932	2.335
2.200	-407.942002	-407.982123	-407.982225	+0.102	40.2	19,966	2.339
2.400	-407.915865	-407.957314	-407.957425	+0.111	41.6	19,881	2.329
2.600	-407.887294	-407.929953	-407.930087	+0.134	42.8	19,881	2.329
2.900	-407.844982	-407.884942	-407.885013	+0.071	40.0	19,880	2.328
<i>3.200</i>	-407.806559	-407.862021	-407.853522	<i>void</i>	—	19,881	2.329
<i>3.600</i>	-407.764023	-407.849591	-407.825353	<i>void</i>	—	19,881	2.329
4.200	-407.719485	-407.840669	-407.840733	+0.065	121.2	19,881	2.329
5.000	-407.686955	-407.837262	-407.837292	+0.030	150.3	19,880	2.328
6.000	-407.668269	-407.836424	-407.836433	+0.009	168.2	11,118	1.302

Across the eleven valid geometries the largest error is 0.134 mHa, a factor of twelve within chemical accuracy, and the smallest is 0.009 mHa. Hartree–Fock errs by 35.5 to 168.2 mHa over the same points.

The derived quantity of Table 8 constitutes a more stringent test than any individual total energy, a dissociation energy being a difference in which the systematic component of the error cancels.

Table 8: Quantities derived from the dissociation curve.

	HI-VQE	CASCI
Minimum on this grid	1.900 Å	1.900 Å
Dissociation energy to the final point	103.89 kcal mol ⁻¹	103.94 kcal mol ⁻¹
Error in the dissociation energy		0.051 kcal mol ⁻¹

4.4 Invalidity of the reference at two geometries

At $r = 3.20$ and 3.60 Å the reported errors are -8.499 and -24.238 mHa. An energy below the reference is not a superior result; in a correct variational calculation it is not attainable. The workflow labels both INCONSISTENT rather than reporting them.

First argument, from the variational bound. The HI-VQE energy is the lowest eigenvalue of PHP , and the sampling circuit commutes with both spin-resolved number operators, so the sampled configurations do not leave the $S_z = 0$ sector. Consequently

$$E_{\text{HI-VQE}} \geq \min_{\psi \in S_z=0} \frac{\langle \psi | H | \psi \rangle}{\langle \psi | \psi \rangle} = E_0(S_z = 0).$$

If $E_{\text{HI-VQE}} < E_{\text{CASCI}}$ then $E_{\text{CASCI}} > E_0(S_z = 0)$, and the reference is not the ground state of the sector it was intended to diagonalise. The inequality admits only this direction, so the conclusion concerns the reference and not the method.

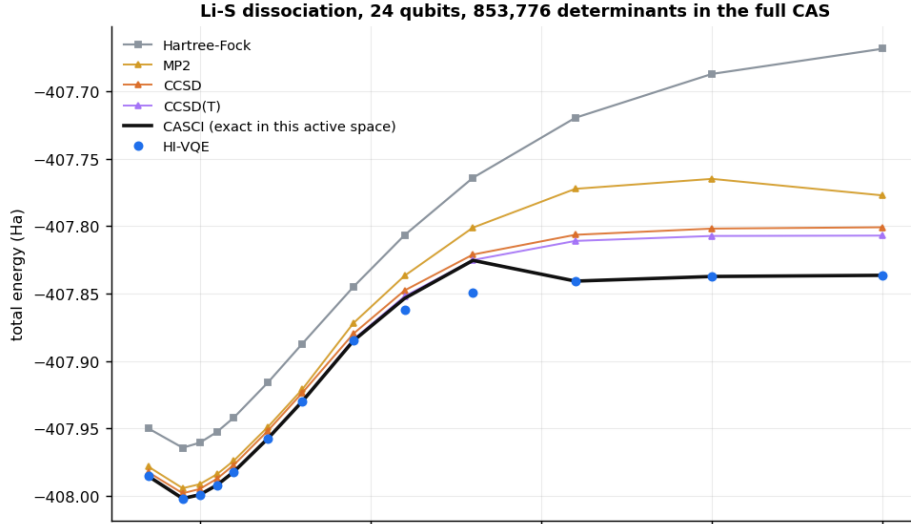


Figure 10: The Li-S dissociation curve, with the classical hierarchy shown against CASCI and HI-VQE points overlaid. The reference curve falls between 3.6 and 4.2 Å; a dissociation curve cannot do so, and this observation constitutes the second argument of Section 4.4.

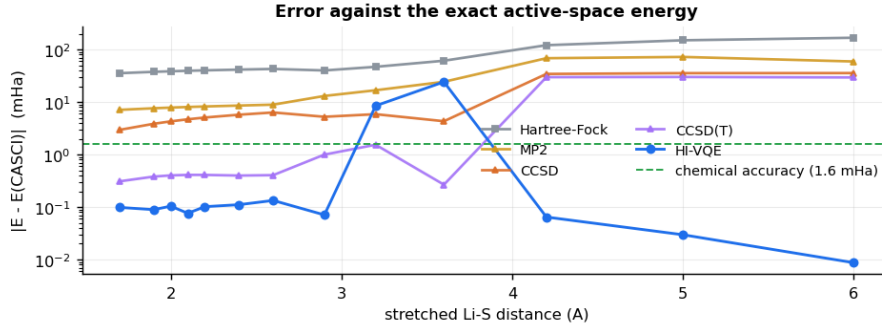


Figure 11: Absolute error against CASCI on a logarithmic scale. Two features determine the remainder of this section: the excursion in the HI-VQE error at 3.2 and 3.6 Å, which reflects failure of the reference rather than of the method (Section 4.4); and the step in the classical curves between 3.6 and 4.2 Å, which coincides with a change of spin sector (Section 4.5).

Second argument, from the shape of the curve. A dissociation curve increases monotonically toward its asymptote. The reference does not:

r (Å)	3.200	3.600	4.200	6.000 (asymptote)
CASCI	-407.853522	-407.825353	-407.840733	-407.836433
HI-VQE	-407.862021	-407.849591	-407.840669	-407.836424

The reference increases from 3.2 to 3.6 Å and then decreases by 15.4 mHa at 4.2 Å, its value at 3.6 Å lying above its own asymptote. The HI-VQE curve increases monotonically to the asymptote throughout. Two independent arguments identify the same two geometries.

Mechanism. PySCF dispatches a closed-shell RHF reference to `direct_spin0`, which exploits the $\alpha \leftrightarrow \beta$ symmetry of the CI matrix. This is a legitimate optimisation and is inappropriate here: a CI vector constrained to be symmetric under that exchange spans only the even- S states, so the solver cannot return a triplet even where the triplet is the ground state. The solver was changed to `direct_spin1`, which imposes only S_z , this being the constraint under which HI-VQE itself operates, and the Davidson convergence was tightened to 10^{-12} , since the near-degeneracy motivating the change also degrades convergence of the final digits.

The correction is incomplete and is reported as such

Following the change the reference correctly returns $\langle \hat{S}^2 \rangle \approx 2$ from 4.20 Å onward. At 3.20 and 3.60 Å it continues to return a singlet. The probable explanation is convergence of the Davidson procedure onto an excited root in a region of near-degeneracy; the diagnostic experiment is a multi-root CASCI calculation at those two geometries, which was not performed. Until it is, both geometries constitute an open defect and all errors there are void, for the classical methods of Table 11 equally as for HI-VQE.

4.5 Spin sector along the coordinate

The diagnostic quantity throughout is $\langle \hat{S}^2 \rangle$, recorded for both the returned state and the reference (Table 9).

Table 9: Spin expectation values along the coordinate. The reference changes sector between 3.60 and 4.20 Å, whereas HI-VQE reaches the triplet from 3.20 Å. The two geometries at which they disagree are precisely the two void geometries.

r (Å)	$\langle \hat{S}^2 \rangle$ (HI-VQE)	$\langle \hat{S}^2 \rangle$ (reference)	Classification
2.900	1.0×10^{-4}	2.5×10^{-14}	CHEMICAL ACCURACY
3.200	2.0000	8.0×10^{-15}	INCONSISTENT
3.600	1.9997	2.1×10^{-14}	INCONSISTENT
4.200	1.9984	2.0000	SPIN CONTAMINATED
5.000	1.9896	1.9850	SPIN CONTAMINATED
6.000	1.0002	1.9467	SPIN CONTAMINATED

The underlying physics is standard. At a dissociated Li-S bond the fragments are $\text{Li}(^2\text{S})$ and $\text{LiS}(^2\Pi)$, two doublets whose singlet and triplet couplings become degenerate; which lies lower is a genuine question whose answer changes along the coordinate. The crossing occurs between 2.90 and 3.20 Å, HI-VQE returning $\langle \hat{S}^2 \rangle \approx 2$ from 3.20 Å onward.

At 6.00 Å the two values differ again, 1.00 against 1.95, yet the energies agree to 0.009 mHa. This combination is the signature of genuine degeneracy: for two non-interacting doublets the singlet and triplet couplings are isoenergetic, so an equal mixture, for which $\langle \hat{S}^2 \rangle = 1$, and a nearly pure triplet are both ground states, and no energy can distinguish them. This is the reason $\langle \hat{S}^2 \rangle$ is reported alongside every energy rather than inferred from it.

The consequence for the classical baselines is the third instance of the failure mode, and the least apparent.

Result 4, stated precisely

MP2, CCSD and CCSD(T) as computed here are constructed on the closed-shell RHF determinant and are therefore singlet-constrained. From 4.20 Å onward the reference against which they are scored has $\langle \hat{S}^2 \rangle \approx 2$. Their reported errors at those geometries are consequently a sum of a spin-sector difference and a correlation-treatment deficiency, and these data cannot separate the two contributions. The step from 0.27 mHa at 3.60 Å to 29.83 mHa at 4.20 Å is indicative precisely because T_1 increases smoothly across the same interval, from 0.0896 to 0.1134: an error increasing by two orders of magnitude while its presumed cause changes by 27% is not attributable to that quantity alone.

One geometry admits the decomposition. At 6.00 Å the singlet and triplet are degenerate, as the 0.009 mHa agreement above demonstrates, so the sector contribution vanishes and the entire 29.48 mHa CCSD(T) error is a correlation-treatment deficiency: the established breakdown of an RHF-based coupled-cluster treatment at dissociation, corroborated by $T_1 = 0.1216$ and by the

fractional occupancies of Figure 7. At 4.20 and 5.00 Å no such argument is available and none is attempted.

4.6 Subspace compression and convergence

The method attains these accuracies within 2.3% of the determinant space, and within 1.3% at 6.00 Å. The strongly correlated geometry requires fewer determinants, since at full dissociation the wavefunction is dominated by a small number of fragment configurations. Both quantities are shown by iteration in Figures 12 and 13.

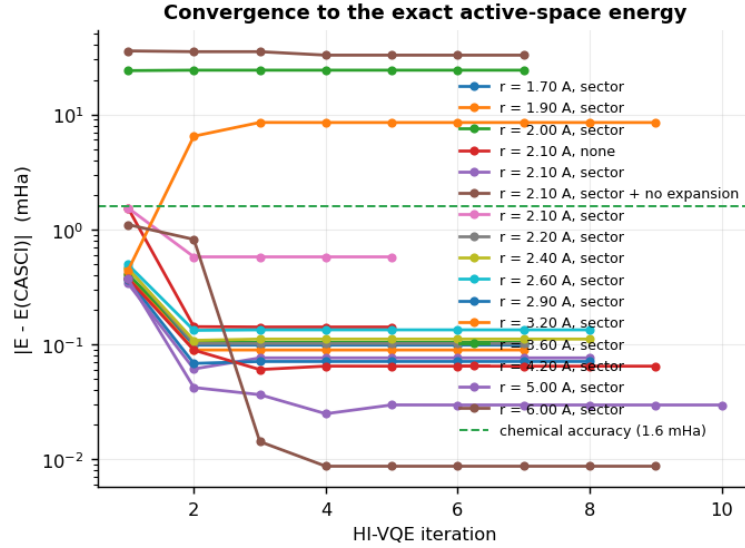


Figure 12: Error against CASCI by iteration for all calculations. Most geometries converge by the fourth iteration and all are stationary by the eighth. The two trajectories plateauing above 8 mHa are the void geometries of Section 4.4; the trajectory near 30 mHa is the no-expansion ablation.

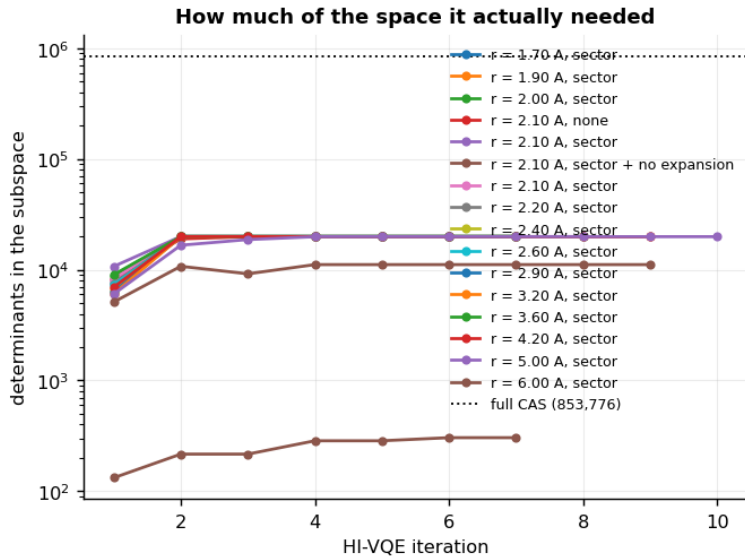


Figure 13: Determinants in the subspace by iteration, against the full CAS dimension of 853,776. The subspace saturates at the 20,000 cap within two iterations for all production calculations; the trajectory near 3×10^2 is the circuit-only ablation, which never populates a usable subspace.

4.7 Ablation of the algorithmic components

Three configurations were computed at $r = 2.10 \text{ \AA}$: the complete method (**sector**); the method with no circuit, the classical expansion proposing every configuration (**none**), which is a conventional selected-CI and constitutes the control of interest; and the circuit alone with the classical expansion removed (**no expansion**).

Table 10: Ablation at $r = 2.10 \text{ \AA}$: identical geometry, Hamiltonian, determinant cap and convergence criteria. One calculation per configuration at seed 1234.

Configuration	Error (mHa)	Determinants	Classification
Complete method (sector)	0.076	19,932	CHEMICAL ACCURACY
Classical control (none)	0.141	19,890	CHEMICAL ACCURACY
Published expansion rule	0.576	19,881	CHEMICAL ACCURACY
Circuit only (no expansion)	32.745	304	SPIN CONTAMINATED

Two statements survive this table and one does not.

Established. The classical expansion is necessary. Its removal costs a factor of 230 in error and reduces the subspace from 19,932 determinants to 304; no plausible seed-to-seed variation spans that interval.

Established. The classical expansion is sufficient at this geometry. The control attains 0.141 mHa, an order of magnitude within chemical accuracy, with no quantum circuit. This is unsurprising given the nature of the classical component, which is a selected-CI procedure; the selected-CI literature [5, 6] reports near-FCI accuracy on spaces substantially larger than 853,776 determinants.

Not established: that the circuit improves upon the control

The complete method’s 0.076 mHa against the control’s 0.141 mHa is a factor of 1.9 obtained from one realisation of each, both stochastic, both at a single seed, and both an order of magnitude within the acceptance threshold. No variance estimate is available and there is accordingly no basis for treating this as a difference. Three to five seeds at this geometry would resolve it.

A figure in the generated output is misleading and is reproduced only to be withdrawn

Figure 14 plots mean error by configuration across all calculations. Its **sector** bar lies at 2.44 mHa and appears inferior to the 0.141 mHa control, which would invert even the weak reading above. This is an artefact: that mean is taken over 14 calculations including the two void geometries of Section 4.4, whose values of -8.5 and -24.2 mHa dominate the average, whereas **none** comprises a single calculation at a single geometry. Averaging over a defective reference does not constitute a comparison. Table 10 supersedes the figure.

4.8 Classical baselines in the same active space

Below the void interval CCSD(T) maintains errors of 0.3 to 1.0 mHa and lies within chemical accuracy at every geometry, HI-VQE being superior by a factor of three to ten, which is a real but not decisive margin. At $6.00 \text{ \AA}$, the one dissociated geometry at which the comparison is unambiguous, CCSD(T) errs by 29.48 mHa, eighteen times chemical accuracy, against 0.009 mHa for HI-VQE, a factor of 3.4×10^3 .

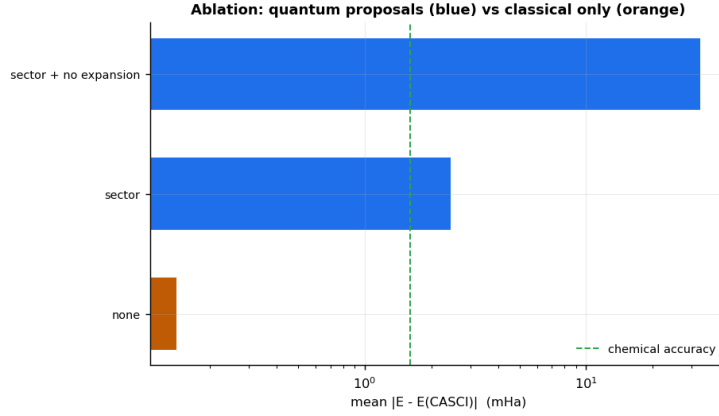


Figure 14: Mean error by ablation configuration, as emitted by the reporting code. This figure is not a result; see the preceding caution. Its **sector** mean is contaminated by the two geometries at which the reference, rather than the method, failed.

Table 11: Classical baselines against CASCI, with the T_1 diagnostic [7]. The rows at 3.200 and 3.600 Å are void for every method (Section 4.4). From 4.200 Å the reference is a triplet while these methods are singlet-constrained, so those errors combine a spin-sector difference with a correlation deficiency; only at 6.000 Å, where the two spin states are degenerate, is the error unambiguously correlation (Section 4.5).

r (Å)	MP2 err (mHa)	CCSD err (mHa)	CCSD(T) err (mHa)	HI-VQE err (mHa)	T_1	Ref. $\langle \hat{S}^2 \rangle$
1.700	+7.11	+2.95	+0.31	+0.099	0.0239	0
1.900	+7.62	+3.86	+0.38	+0.089	0.0270	0
2.000	+7.85	+4.29	+0.40	+0.104	0.0287	0
2.100	+8.06	+4.70	+0.41	+0.076	0.0304	0
2.200	+8.25	+5.08	+0.41	+0.102	0.0321	0
2.400	+8.58	+5.75	+0.40	+0.111	0.0354	0
2.600	+8.93	+6.34	+0.41	+0.134	0.0387	0
2.900	+13.08	+5.24	+1.00	+0.071	0.0509	0
3.200	<i>void</i>	<i>void</i>	<i>void</i>	<i>void</i>	0.0625	0
3.600	<i>void</i>	<i>void</i>	<i>void</i>	<i>void</i>	0.0896	0
4.200	+68.50	+34.37	+29.83	+0.065	0.1134	2.000
5.000	+72.45	+35.54	+30.06	+0.030	0.1197	1.985
6.000	+59.30	+35.67	+29.48	+0.009	0.1216	1.947

Established by Case B

At 6.00 Å, where the reference is verified, the spin states are degenerate and the comparison is therefore unambiguous, HI-VQE reproduces the exact active-space energy to 0.009 mHa while sampling 1.3% of the determinant space, whereas the best classical method available in the same space errs by 29.48 mHa. This is a genuine three-order-of-magnitude result against coupled cluster and is the strongest single quantity reported here.

Not established: that the quantum component is responsible

The comparison above is between HI-VQE and coupled cluster, not between HI-VQE and its own classical component. The `none` control was computed at 2.10 Å only, where it attained chemical accuracy unaided. Since that control is a selected-CI procedure, and selected-CI is by construction a multireference method, the reasonable expectation is that it also performs well at 6.00 Å, possibly as well as the complete method. Were that so, the correct conclusion would be that a classical selected-CI surpasses CCSD(T) by three orders of magnitude at dissociation and that the circuit contributes little, which is a materially different result. Computation of `none` at 4.20, 5.00 and 6.00 Å would decide the matter. It requires no PySCF, only the cached Hamiltonians, and is the single most valuable experiment not performed.

4.9 Comparison with the published study

The source study [1] fixes the active space, the reference method, the algorithm and the nature of the coordinate. It does not report a geometry, per-geometry energies or an orbital set, and does not specify the sampling circuit. Total energies are therefore not reproducible from it and no such claim is made. The comparable quantities, and those measured here, are the error against the present CASCI reference at each geometry, the fraction of the determinant space required, and the shape of the curve.

5 Discussion

5.1 A common failure mode

The three defects described above appear unrelated: a comparison of ansätze, a reference energy, and a set of classical baselines. They are the same defect.

Table 12: One failure mode in three instances. In each, a plausible quantity was produced by comparing states that were not comparable, and in each the diagnostic that identified it was inexpensive.

Location	Quantities compared	Appearance	Diagnostic
Case A, §3.5	Two operator pools, the calculations coinciding for 31 iterations and differing only in budget	A pool comparison, 0.892 against 0.017 mHa	Comparison of trajectories rather than endpoints
Case B, §4.4	A CASCI reference against a variationally bounded method in the same sector	A method error of −24 mHa	Sign of the error; monotonicity of the curve
Case B, §4.5	Singlet-constrained coupled cluster against a triplet reference	A correlation discontinuity between 3.6 and 4.2 Å	Recording $\langle \hat{S}^2 \rangle$ of the reference

The common structure is that each produced a quantity of the expected magnitude. None resembled a numerical fault. The first two would have been reported as results; the third would have been reported as the principal finding.

5.2 A variationally bounded method as a certificate on its reference

The detection described in Section 4.4 was not fortunate. It is an instance of a property that any variationally bounded method possesses, and which is worth stating in general form because it inverts the customary relationship between a quantum calculation and the classical reference used to judge it.

Let $\mathcal{S}$ denote a symmetry sector, here $S_z = 0$, and let a method return $E_q = \min \text{spec}(PHP)$ for some projector P onto a subspace of $\mathcal{S}$. Then $E_q \geq E_0(\mathcal{S})$ for every P , with no assumption whatever on how P was obtained. Consequently, for any reference energy E_{ref} asserted to equal $E_0(\mathcal{S})$,

$$E_q < E_{\text{ref}} \implies E_{\text{ref}} > E_0(\mathcal{S}),$$

that is, the assertion is false. The implication runs in one direction only, which is precisely what makes it useful: a violation is not evidence of a superior quantum result and cannot be produced by sampling noise, an underconverged circuit or a poor choice of subspace, since all of these can only raise E_q . The only remaining explanation is that the reference is not the sector ground state.

Three features make this practical. The test costs nothing beyond the sign of a quantity already computed. It requires no knowledge of the exact answer, which is the regime in which references are least trustworthy. And it is not specific to HI-VQE: sample-based diagonalisation and classical selected-CI share the same bound, so the same certificate is available to any of them.

The customary hierarchy places the classical reference in judgement over the quantum result. Where the quantum method carries a variational bound and the reference does not carry a proof of convergence, the direction of inference is the reverse of the customary one. In this study the CASCI reference at 3.20 and 3.60 Å was produced by a solver whose default restricted it to even- S states, and no diagnostic internal to that calculation reported a difficulty: it converged, and returned a plausible energy. What identified the defect was a quantum calculation being scored against it.

A recommended check

Where a reference is obtained from an iterative eigensolver and a variationally bounded method is available in the same sector, the sign of the error is a convergence diagnostic for the reference and should be recorded as such. Sustained negative errors indicate a reference defect and not a method success. The corresponding positive statement is weaker but still useful: a positive error confirms only that the reference lies below the method, which is consistent with, but does not establish, convergence of the reference.

5.3 Interpretation of the correlation diagnostic

Table 11 might be read as licensing a decision rule: evaluate T_1 , and where it exceeds a threshold, adopt a quantum method. These data support the first component of that rule and not the second. What T_1 predicts here is the point at which coupled cluster ceases to be adequate, which is well established and with which these data are consistent. What it does not predict is that a quantum method is thereby required, since the classical alternative in the relevant regime is not coupled cluster but selected-CI, itself a multireference method, which the `none` control shows attaining chemical accuracy unaided at the one geometry where it was computed.

The defensible form of the crossover statement

Beyond the multireference crossover, single-reference coupled cluster fails by tens of millihartree while HI-VQE does not. Whether the quantum component of HI-VQE is necessary for this, or whether its classical selected-CI component suffices, is not determined by these data; at the one geometry where the control exists, the classical component sufficed.

5.4 Cost and yield of the additional instrumentation

The additional instrumentation comprised one evaluation of $\langle \hat{S}^2 \rangle$ per calculation for the returned state, one for the reference, a validation receipt per Hamiltonian, and three SHA-256 digests: approximately one second of computation per Hamiltonian and a few hundred lines of code.

It yielded the three findings of Table 12 together with identification of a superseded dataset otherwise indistinguishable from the current one. Every substantive result reported here derives from the instrumentation rather than from the energies, which was not the anticipated outcome.

6 Claims not supported by these data

Each claim declined is listed below with the experiment that would decide it and the expected outcome. The expectations are recorded so that they may be falsified.

1. **That the quantum sampler is necessary at dissociation.** The none control exists at 2.10 Å only, where it attained 0.141 mHa without a circuit. *Experiment:* compute none at 4.20, 5.00 and 6.00 Å; no PySCF is required, the Hamiltonians being cached. *Expectation:* the control remains within chemical accuracy at all three, and the margin of the complete method over it remains below a small factor. Were that so, the strongest quantity reported here (Section 4.8) would be a result concerning selected-CI rather than quantum sampling.
2. **That the ordering of Case A survives into a strongly correlated system.** ADAPT-VQE was computed only on imipramine, where CCSD(T) surpassed it, and the Li₂S coordinate, where CCSD(T) fails, was treated only with HI-VQE. The two cases therefore never meet. *Experiment:* ADAPT-VQE at 2.10 and 6.00 Å on the cached Li₂S Hamiltonian; the implementation already exists in the workflow. *Expectation:* the ordering reverses, ADAPT-VQE surpassing CCSD(T) at 6.00 Å while remaining behind it at 2.10 Å. This would convert the two isolated cases into a single controlled comparison across the correlation crossover, and is the experiment that would most improve the study as a whole.
3. **That the complete method surpasses its own classical component.** The values 0.076 and 0.141 mHa are single realisations without variance estimates, both an order of magnitude within the threshold. *Experiment:* three to five seeds per configuration at 2.10 and 6.00 Å. *Expectation:* the intervals overlap.
4. **Any conclusion regarding UCCSD relative to UCCGSD.** The calculations coincide for 31 iterations and differ in budget (Section 3.5). *Experiment:* repeat the UCCSD calculation with a 100-operator budget. *Expectation:* convergence to within a few hundredths of a millihartree of Calculation 2.
5. **Any quantity at 3.20 and 3.60 Å.** The reference there is not the sector ground state (Section 4.4). *Experiment:* CASCI on an ROHF triplet reference at both geometries, which is simpler than the multi-root calculation and returns the same information. *Expectation:* a triplet root at or below the HI-VQE energies, −407.862021 and −407.849591 Ha, restoring monotonicity of the reference curve and validating the certificate of Section 5.2 against an independent calculation.
6. **Decomposition of the CCSD(T) error at 4.20 and 5.00 Å.** Those errors combine a spin-sector difference with a correlation deficiency (Section 4.5). *Experiment:* ROHF-based UCCSD(T) on the triplet at both geometries. *Expectation:* a substantial reduction in error that nonetheless remains well outside chemical accuracy, $T_1 > 0.11$ indicating a genuine multireference problem in addition.

7. **That the ordering of Case A persists into larger active spaces.** CAS(6e,6o) comprises 400 determinants (Section 3.7), and no experiment in this study bears upon the extrapolation. Item 2 addresses the correlation regime but not the size of the active space, which would require a system beyond the reach of exact diagonalisation and therefore beyond the verification strategy adopted here.

Items 1 to 3 require hours of computation on hardware already available and would alter what this study is able to claim. They are the recommended next step, in preference to any additional system. Item 1 could invert the strongest quantity reported here; item 2 would join the two cases into a single controlled comparison; item 3 would supply the variance estimates that several of the present statements lack.

A subsequent extension would add one further stage. The question these data cannot reach, whether any of this survives where exact diagonalisation is unavailable, requires a system large enough to be of interest while retaining a published reference. The 36-qubit methane dimer of Kaliakin *et al.* [8], benchmarked against CASSI in the same active space, is the natural candidate, and the failure modes catalogued in Table 12 are those for which instrumentation should be prepared in advance.

7 Limitations

Absence of hardware results. All calculations are CPU simulations. The imipramine ansatz requires 2988 two-qubit gates at depth 5296; the Li₂S sampling circuit requires 992 CZ gates at depth 451 on a linear chain. Neither returns the accuracy reported here on any presently available device.

Single random seed. Every HI-VQE calculation employs seed 1234, 8192 shots and SPSA optimisation. No result depending on a difference smaller than an order of magnitude should be regarded as established.

Two void geometries. The geometries at 3.20 and 3.60 Å are excluded from all summary statistics and are labelled rather than omitted, but they constitute a defect.

Single conformer and single coordinate. Case A employs one PubChem conformer. Case B extends one bond in a linear geometry; the physical dissociation path is not constrained to be linear.

Small active spaces. CAS(6e,6o) and CAS(12e,12o) were selected because exact diagonalisation is feasible within them. Neither is large enough for the active-space error, the difference between the exact active-space result and the exact result for the molecule, to be small. Every accuracy claim reported here is relative to the active space and bears no implication for accuracy relative to experiment.

Indicative timings. Wall times were obtained on shared cloud instances across two sessions with differing library versions (Appendix C). They support statements of order of magnitude and no finer claim.

Reproduction of methodology, not of total energies. Neither source publication reports the geometry or orbital set required to reproduce its total energies. Methodology and error metrics alone are reproduced.

8 Conclusions

Two published quantum algorithms were executed on the active spaces specified by their source publications, every Hamiltonian was verified against exact diagonalisation with a recorded receipt, and the spin state of both the returned wavefunction and the reference was instrumented.

The numerical results stand. ADAPT-VQE converges to 0.017 mHa on imipramine’s 12-qubit active space, where CCSD(T) attains 0.008 mHa and the comparison constitutes a calibration rather than a contest. HI-VQE reproduces exact diagonalisation to better than 0.15 mHa at eleven of thirteen Li_2S geometries while sampling 2.3% of the determinant space and requiring one measurement setting in place of 15,697; at 6.00 Å, the single dissociated geometry at which the reference is verified and the spin states are degenerate, it attains 0.009 mHa against 29.48 mHa for CCSD(T).

One quantum-side advantage is established without qualification. The reduction in measurement cost, from 15,697 Pauli words per energy evaluation to a single computational-basis setting, is structural. It does not depend on the random seed, on the reference energy, or on any spin sector, and it is therefore untouched by every comparison questioned below. Where the surviving accuracy claims are narrow, this one is not.

The unanticipated result is that three of the study’s comparisons were invalid for a common reason, and that each was identified by recording one additional expectation value: a comparison of ansätze in which the ansätze coincided; a reference that was not the ground state of its own symmetry sector; and classical baselines scored across a change of spin sector. None resembled a numerical fault, and each produced a quantity of the magnitude a reader would anticipate.

The second of these generalises, and is the most transferable outcome of the study. A method returning a rigorous upper bound on the ground state of a symmetry sector certifies any reference asserted to be that ground state: an energy below the reference falsifies the reference and cannot be produced by sampling noise, an underconverged circuit or a poor subspace, all of which raise the bound. The customary direction of judgement is thereby reversed, the quantum calculation auditing the classical reference against which it is scored. Here the defective CASCI energies reported no internal difficulty – the solver converged and returned plausible values – and were identified only because a variationally bounded method was being scored against them. The check costs the sign of a quantity already computed and is available to sample-based diagonalisation and to classical selected-CI on the same terms.

The surviving claims are therefore reported narrowly, and Section 6 records the six declined together with the experiments that would decide them. Three of those experiments require hours of computation on hardware already available: one could invert the strongest quantity reported here, one would join the two cases into a single controlled comparison across the correlation crossover, and one would supply the variance estimates that several present statements lack. That none has been performed is the most consequential statement in this document.

A Provenance digests

Every result is addressed by the digests of Table 13. The prefixes are the first twelve hexadecimal characters of the SHA-256 digests recorded in each artefact; complete digests are contained in the data files.

Table 13: Provenance digests. **physics** covers only those modules capable of altering a Hamiltonian; **workflow** covers every module in the directory. Digests are computed over line-ending-normalised file bytes.

Artefact	spec	physics	workflow
Imipramine Hamiltonian and receipt	b85747c4a609	c6cd75127aac	c4682e8536f6
Imipramine ADAPT calculations	b85747c4a609	c6cd75127aac	afdc411eda45
Li_2S cache, 13 geometries	3c9616b7ad31	f6989a3a6000	172e18c71130
Li_2S HI-VQE results	3c9616b7ad31	f6989a3a6000	172e18c71130
<i>Superseded; not used for any quantity reported here:</i>			
Li_2S cache, pre-correction	3c9616b7ad31	d3a3a1ffa0a9	5e074369afe5

Two observations follow from the table and are recorded because they are not otherwise visible. The imipramine ADAPT calculations carry a `workflow` digest, `afdc411eda45`, differing from that under which the Hamiltonian was constructed and validated, `c4682e8536f6`, while `physics` is identical. A module incapable of altering the Hamiltonian was modified between the two operations. The energies are therefore attached to the correct Hamiltonian, and a byte-exact repetition of the trajectory is not guaranteed from the current source tree.

The superseded `Li2S` cache differs in `physics`, the digest governing scientific comparability. That difference is the `direct_spin0` to `direct_spin1` change described in Section 4.4.

B Complete settings

B.1 Case A, ADAPT-VQE

Setting	Calculation 1	Calculation 2
Operator pool	UCCSD (63)	UCCGSD (435)
Operator budget	31	100
Gradient tolerance	10^{-3}	10^{-3}
Initial state	HF determinant, $\theta = 0$	HF determinant, $\theta = 0$
Number penalty	0	0
Spin penalty	0	0
Contamination price	1 Ha per unit $\langle \hat{S}^2 \rangle$	1 Ha per unit $\langle \hat{S}^2 \rangle$
Backend	exact sparse statevector	exact sparse statevector
Circuit synthesis	exact, Qiskit, level 3	exact, Qiskit, level 3
Hamiltonian cutoff	10^{-6} Ha, 0 discarded	10^{-6} Ha, 0 discarded

B.2 Case B, HI-VQE

Setting	Value
Simulator	<code>sector</code>
Maximum determinants	20,000
Classical expansion	400 candidates per iteration
Expansion references	8 (the published rule is 1; computed separately)
Amplitude threshold	10^{-6}
Ranking	second-order perturbative
Shots	8192
Circuit repetitions	1
Optimiser	SPSA, $a = 0.25$, $c = 0.1$, one step per iteration
Convergence	10^{-6} Ha sustained over 5 iterations
Maximum iterations	12
Random seed	1234 (single seed; see Section 6)
Readout and depolarising error	0 and 0
Circuit qubits	24
Circuit depth	29 (451 transpiled to a linear chain)
Two-qubit gates	298 (992 CZ transpiled)
Parameters	322

C Software environment

The two studies were executed in separate sessions and the package versions differ. Both are recorded rather than harmonised, a version difference being a candidate explanation for any subsequent discrepancy.

Package	Case A (imipramine)	Case B (Li ₂ S)
Python	3.12.13	3.12.13
NumPy	1.26.4	2.5.2
SciPy	1.17.1	1.16.3
PySCF	2.14.0	2.14.0
Qiskit	1.0.2	2.5.2
Qiskit Aer	0.14.2	0.17.2
OpenFermion	1.8.1	1.8.1
Platform	Linux 6.6.122	Linux 6.6.122, glibc 2.35

D Reproduction

Both workflows present a single command interface. PySCF is required only for the `prepare` and `reference` stages; the quantum stages read the cached Hamiltonian and execute anywhere, which is why the experiments of Section 6 are inexpensive.

Stage	Command
<i>Case A, imipramine</i>	
Verify the installation	<code>python run.py selftest</code>
Construct the Hamiltonian	<code>python run.py prepare</code>
Validate it	<code>python run.py validate</code>
Classical baselines	<code>python run.py classical</code>
ADAPT-VQE	<code>python run.py adapt -pool uccgsd</code>
<i>Case B, Li₂S</i>	
Verify the installation	<code>python run.py selftest</code>
Construct the Hamiltonian	<code>python run.py prepare -distance 2.10</code>
Validate it	<code>python run.py validate -distance 2.10</code>
HI-VQE	<code>python run.py hivqe -distance 2.10</code>
Generate the report	<code>python run.py report</code>

Verification of a digest requires prior normalisation of line endings (Section 2.2).

E Data inventory

Superseded material is listed explicitly rather than deleted, the existence of a discarded dataset being itself a result (Section 2.2).

Artefact	Status	Note
Imipramine Hamiltonian and receipt	used	Digests match the current source tree exactly.
Imipramine <code>adapt_uccsd</code>	used	Truncated calculation; Table 4.
Imipramine <code>adapt_uccgsd</code>	used	Converged calculation; Table 4.
Imipramine <code>classical</code>	used	MP2, CCSD and CCSD(T). Its <code>reference_energy</code> field is null; the exact reference is carried in the validation receipt.
Li ₂ S cache, 13 geometries	used	<code>physics f6989a3a6000</code> .
Li ₂ S results, 16 calculations	used	<code>workflow 172e18c71130</code> .
Li ₂ S cache, 3 geometries	superseded	<code>physics d3a3a1ffa0a9</code> , preceding the <code>direct_spin1</code> correction. CASCI at 6.00 Å differs by 10.1 mHa and RHF by 23.9 mHa from the current set.
Li ₂ S results, 6 calculations	superseded	Constructed on the above. Its dissociation energy is in error by 1.586 kcal mol ⁻¹ against 0.051 for the current set, which is the consequence of the <code>direct_spin0</code> defect.

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